

Brain rust: recent discoveries on the role of oxidative stress in neurodegenerative diseases.



Oxidative stress (OS) and damages due to excessive reactive oxygen species (ROS) are common causes of injuries to cells and organisms. The prevalence of neurodegenerative diseases (ND) increases with aging and much of the research involving ROS and OS has emerged from works in this field. This text reviews some recent published articles about the role of OS in ND. Since there are many reviews in this field, the focus was centered in articles published recently. The Scientific Journals Directory supported by the Brazilian Ministry of Education Office for the Coordination of Higher Educational Personnel Improvement (CAPES) was used to search, download, and review articles. The search engine looked for the terms 'oxidative stress AND neurodegenerative diseases AND nutrition' in 10 different scientific collections. Biochemical markers for ND lack sensitivity or specificity for diagnosis or for tracking response to therapy today. OS has an intimate connection with ND, albeit low levels of ROS seem to protect the brain. Deleterious changes in mitochondria, OS, calcium, glucocorticoids, inflammation, trace metals, insulin, cell cycle, protein aggregation, and hundreds to thousands of genes occur in ND. The interaction of genes with their environment, may explain ND. Although OS has received much attention over the years, which increased the number of scientific works on antioxidant interventions, no one knows how to stop or delay ND at present. Interventions in vitro, in vivo, and in humans will continue to contribute for a better understanding of these pathologies.